



CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Strategic insights from 3,900 transactions to drive business growth and customer engagement

PROJECT SCOPE & OBJECTIVES

THE CHALLENGE

Uncover hidden patterns in customer behavior using transactional data from 3,900 purchases across diverse product categories. Our goal: transform raw data into actionable business strategies.

STRATEGIC FOCUS

- Identify high-value customer segments
- Analyze spending patterns and preferences
- Measure subscription impact on revenue
- Optimize discount and promotional strategies



DATASET FOUNDATION

3.9K

TRANSACTION RECORDS

Comprehensive purchase data spanning multiple categories and seasons

18

DATA FIELDS

Rich customer demographics, purchase details, and behavioral metrics

37

MISSING VALUES

Minimal missing data in review ratings, imputed using category medians

KEY DATA DIMENSIONS

CUSTOMER PROFILE

- Age and gender distribution
- Geographic location
- Subscription status

PURCHASE DETAILS

- Product category and item
- Purchase amount and season
- Size and color preferences

BEHAVIORAL INSIGHTS

- Discount and promo usage
- Shipping preferences
- Review ratings

DATA PREPARATION PIPELINE



DATA LOADING

Import dataset using pandas and validate structure with `df.info()` and `describe()` methods



CLEANING & IMPUTATION

Handle missing review ratings by imputing median values per product category



STANDARDIZATION

Rename columns to snake case for consistency and better documentation practices



FEATURE ENGINEERING

Create age groups and purchase frequency calculations from raw data



DATABASE INTEGRATION

Connect to PostgreSQL and load cleaned DataFrame for SQL analysis

KEY BUSINESS INSIGHTS



REVENUE BY GENDER

Male customers generated **\$157,890** in revenue versus **\$75,191** from female customers



HIGH-VALUE DISCOUNT USERS

839 customers used discounts while spending above average purchase amount of \$59.76



TOP-RATED PRODUCTS

Gloves lead with 3.86 average rating, followed by Sandals (3.84) and Boots (3.82)





SHIPPING & SUBSCRIPTION ANALYSIS

SHIPPING TYPE IMPACT

- **Standard:** \$58.46 average purchase
- **Express:** \$60.48 average purchase

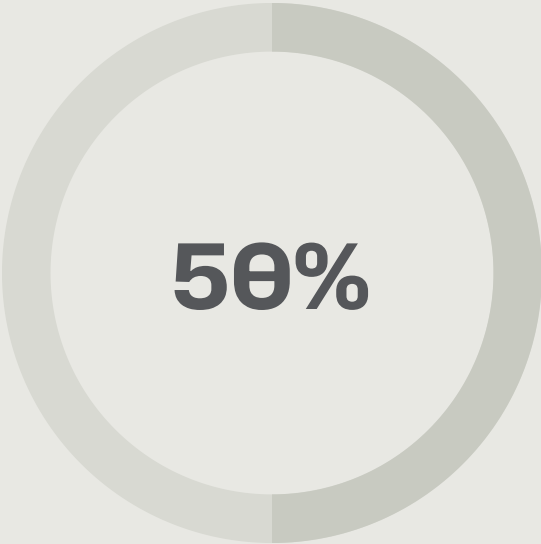
Express shipping correlates with 3.5% higher average spend, suggesting premium customer segment.

SUBSCRIPTION PERFORMANCE

- **1,053 subscribers** generating \$62,645 revenue
- **2,847 non-subscribers** generating \$170,436 revenue

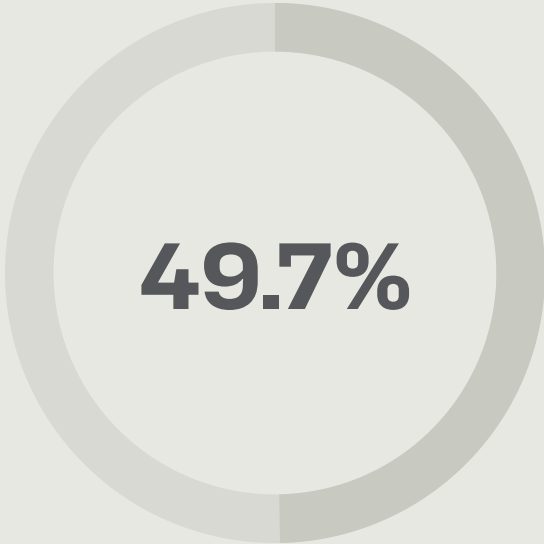
Average spend nearly identical (\$59.49 vs \$59.87), indicating subscription doesn't drive higher purchase amounts.

PRODUCT PERFORMANCE DEEP DIVE



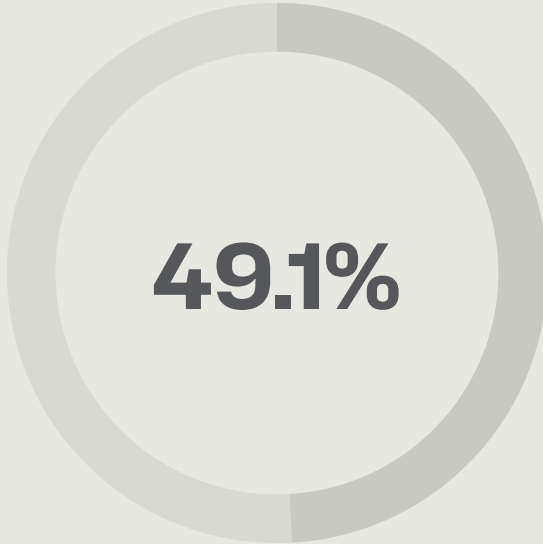
HATS DISCOUNTED

Half of all hat purchases received discounts



SNEAKERS DISCOUNTED

Nearly half of sneaker purchases discounted



COATS DISCOUNTED

High-ticket outerwear frequently discounted

BEST-SELLING PRODUCTS BY CATEGORY

ACCESSORIES

Jewelry: 171 orders
Sunglasses: 161 orders
Belt: 161 orders

CLOTHING

Blouse: 171 orders
Pants: 171 orders
Shirt: 169 orders

FOOTWEAR

Sandals: 160 orders
Shoes: 150 orders
Sneakers: 145 orders

OUTERWEAR

Jacket: 163 orders
Coat: 161 orders

CUSTOMER SEGMENTATION



NEW CUSTOMERS

83 customers with first-time purchases requiring engagement strategies



RETURNING CUSTOMERS

701 customers with 2-5 purchases eligible for loyalty program conversion



LOYAL CUSTOMERS

3,116 customers with 5+ purchases representing core revenue base

AGE GROUP REVENUE DISTRIBUTION

- **Young Adult:** \$62,143 total revenue
- **Middle-aged:** \$59,197 total revenue
- **Adult:** \$55,978 total revenue
- **Senior:** \$55,763 total revenue

STRATEGIC RECOMMENDATIONS

01

BOOST SUBSCRIPTION VALUE

Develop exclusive benefits that demonstrably increase average spend to justify subscription fees

02

LAUNCH LOYALTY PROGRAMS

Reward repeat buyers with tiered incentives to accelerate movement into loyal customer segment

03

OPTIMIZE DISCOUNT STRATEGY

Balance sales volume increases with margin preservation, particularly for high-discount products

04

ENHANCE PRODUCT POSITIONING

Feature top-rated and best-selling products in campaigns to leverage social proof and proven demand

05

IMPLEMENT TARGETED MARKETING

Focus high-revenue age groups and express-shipping customers with personalized messaging



EXECUTIVE SUMMARY

KEY FINDINGS

Male customers drive 68% of total revenue while female customers show higher purchase frequency. Loyal customers represent 79% of the base with strong repeat purchase behavior. Express shipping correlates with premium spending patterns.

REVENUE OPPORTUNITIES

Convert returning customers to loyal segment through targeted incentives. Optimize discount strategy on high-margin products. Develop subscription benefits that demonstrably increase customer lifetime value.